

CPSC 250

Plotting Error Bands

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1 Motivation

When we fit a curve to data, the fit parameters have uncertainties.

A natural question is:

How do uncertainties in the fit parameters translate into uncertainties in the fit itself?

The full statistical treatment is complicated.

However, we can create a useful visualization called an **error band**.

2 The Basic Idea

Suppose our fit parameters are stored in:

- `popt` — optimal fit parameters,
- `pcov` — covariance matrix.

The covariance matrix describes the uncertainties and correlations of the parameters.

We can use this information to generate many random possible fits.

3 Monte Carlo Sampling

The key idea is:

1. randomly generate many possible parameter sets,
2. compute a fit curve for each parameter set,
3. examine the spread of the curves.

4 Generating Random Parameter Sets

We use:

```
ps = np.random.multivariate_normal(  
    popt,  
    pcov,  
    10000  
)
```

This generates:

- 10,000 random parameter sets,
- sampled from a multivariate Gaussian distribution,
- centered on the best-fit parameters.

5 Why a Multivariate Distribution?

Fit parameters are often correlated.

For example:

- increasing one parameter,
- while decreasing another,

might produce similarly good fits.

The covariance matrix captures these correlations.

6 Generating Many Curves

Next:

```
ysample = np.asarray(  
    [  
        fitfunction(xfit, *pi)  
        for pi in ps  
    ]  
)
```

This creates:

- 10,000 fitted curves,
- one for each sampled parameter set.

7 Understanding the Shape of ysample

The array:

```
ysample
```

contains many different possible y-values for each x-position.
Each row corresponds to a different sampled fit.

8 Constructing the Error Band

We now compute percentiles.

Upper boundary:

```
upper = np.percentile(
    ysample,
    84.0,
    axis=0
)
```

Lower boundary:

```
lower = np.percentile(
    ysample,
    16.0,
    axis=0
)
```

9 Why 16% and 84%?

For a Gaussian distribution:

- about 68% of values lie within one standard deviation,
- leaving roughly 16% in each tail.

Therefore:

- the 16th percentile corresponds approximately to -1σ ,
- the 84th percentile corresponds approximately to $+1\sigma$.

10 Plotting the Band

We can shade the region between the boundaries:

```
plt.fill_between(
    xfit,
    lower,
    upper,
    alpha=0.3
)
```

This creates a shaded uncertainty region around the fit.

11 Interpretation

The error band gives a visual estimate of:

- how uncertain the fit is,
- where predictions are more reliable,
- where uncertainties become large.

12 Important Note

This procedure assumes:

- approximately Gaussian parameter uncertainties,
- a reasonable covariance matrix,
- a good fit model.

In professional data analysis, uncertainty estimation can become much more sophisticated.

13 Complete Workflow

Typical steps:

1. Fit the data.
2. Obtain `popt` and `pcov`.
3. Generate many random parameter sets.
4. Compute many fitted curves.
5. Determine percentile boundaries.
6. Plot the shaded region.

14 Key Takeaways

- Fit parameters have uncertainties.
- The covariance matrix describes parameter uncertainty and correlation.
- Monte Carlo sampling can propagate uncertainties into the fitted curve.
- Error bands provide a useful visualization of fit uncertainty.